

Timur Khairulov

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EDUCATION

2024 - Present	MS in Artificial Intelligence at Chung-Ang University CAYSS scholarship recipient	(GPA: 4.22/4.5)
2018 - 2023	BS in Computer Science at Electrotechnical University "LETI"	(GPA: 4.3/5)
2022 - 2023	Exchange Semester at Inha University	(GPA: 3.9/4.5)

WORK EXPERIENCE

C++ Software Developer	LTE L1, YADRO	May 2023 - February 2024
– Developed real-time signal processing modules for 4G base stations using C++ and AVX SIMD instructions, improving throughput by 30% through optimized vectorization – Optimized memory-intensive algorithms in telecommunications software, reducing latency by 25% through cache-aware programming and efficient data structure design – Led development of L1-L2 communication visualization application with team of 3 developers, enabling project managers to analyze telecommunication delays using Python, TypeScript, and MongoDB – Collaborated with cross-functional team of 8 engineers to deliver production-ready code meeting strict real-time performance requirements for deployed base stations		

TEACHING EXPERIENCE

Teaching Assistant, Computer Vision, Chung-Ang University	Spring 2025
– Prepared course materials for lecture on CNN architectures and contributed to exam preparation and solution evaluation – Assisted professor with grading and providing feedback on student assignments and exam solutions	

PROJECTS

VLM Compression: Optimization Techniques for Vision-Language Models	GitHub Blog
<i>Python, PyTorch, BitsAndBytes</i> December 2025	
– Benchmarked 8-bit and 4-bit quantization performance for BLIP-2 and LLaVA models – Implemented and compared structural pruning of MLP and key-value-query components in language model – Uploaded pruned models to HuggingFace with comprehensive analysis published on blog	
Curriculum Learning Fine-Tuning for Mathematical Reasoning	GitHub Blog
<i>Python, PyTorch, Transformers, LoRA, HuggingFace</i> November 2025	
– Fine-tuned SmoLLM2 and Phi-3 models on GSM8K mathematical reasoning dataset using complexity-based curriculum learning strategies – Achieved 1.37% accuracy improvement on difficult problems compared to baseline random-order training – Published comprehensive analysis with reproducible code and demo	
BAGen: Background Animation Generation for Children's Songs	GitHub
<i>Python, PyTorch, Diffusion Models</i> January 2025	

- Built end-to-end pipeline generating synchronized lyrics and animated backgrounds for children’s songs using LLMs and diffusion models
- Developed production-ready system with GUI and CLI interfaces
- Submitted for publication in scientific reports

Information-Theoretic Feature Selection Library

MATLAB, Python | August 2024

GitHub

- Implemented and configured 30+ unsupervised feature selection algorithms with 25 curated datasets
- Created extensible experimental framework for testing novel joint entropy maximization method
- Developed as foundation for Master’s thesis research and future journal publication

PUBLICATIONS

Journal Articles

1. ChainImputer: A Neural Network-Based Iterative Imputation Method Using Cumulative Features
MDPI Symmetry | 2025 | DOI: 10.3390/sym17060869
2. EP-REx: Evidence-Preserving Receptive-Field Expansion for Efficient Crack Segmentation
MDPI Symmetry | 2025 | DOI: 10.3390/sym17101653

Conference Papers

1. Diffusion Model-Based Generative Pipeline for Children Song Video
IEEE International Conference on Consumer Electronics (ICCE) | 2025 | IEEE Xplore
2. Review of current approaches in the area of object detection for autonomous vehicles
Institute of Electronics and Information Engineers (IEIE) | 2024 | DBPIA

TECHNICAL SKILLS

Programming	C/C++, Python, MATLAB, Dart, SQL
Machine Learning	PyTorch, TensorFlow, NumPy, Pandas, Scikit-learn, OpenCV, HuggingFace Transformers
Development	Git, Linux, CMake, Docker, Qt, Flutter, Weights & Biases
Systems	Parallel Programming (AVX), Operating Systems, Computer Architecture
Languages	English (IH - OPIc Certificate), Russian (Native), Korean (Basic)

AWARDS AND HONORS

- 2024 - Present
- CAYSS Scholarship Recipient at Chung-Ang University
- 2018 - 2023
- Bachelor’s Degree Scholarship Recipient at Electrotechnical University ”LETI”